

MBM University Jodhpur
Department of Mechanical Engineering



5ME43B(ME)
Production Engineering Lab III
Third year/ 5thSemester

Machine Tool Metrology

Introduction

- In mass production accuracy is paramount for universal assembly
- The accurate production of the component parts depends upon the accuracy of the machine tools.
- The quality of work-pieces depends upon :
 - Stiffness and rigidity of the machine tool and its components parts: Depend on designer and once tests on a prototype give satisfactory results, there is no need to test the machine of same design over and over again.
 - The alignment of various machine parts in relation to one another.
 - The quality and accuracy of the control devices and the driving mechanism.
- Alignment tests are carried out to check the grade of manufacturing accuracy of the machine tool.

Introduction (Contd.)

- The accuracy of machine tools is tested by means of:
 - Geometrical Test/ Alignment tests:
 - Check the grade of manufacturing accuracy of machine tool
 - Involves checking the relationship between various machine elements when the machine tool is idle.
 - Performance Tests/Practical Tests
 - Check the accuracy of the finished component.
 - It consists of preparing the actual test jobs on the machine and checking the accuracy of the jobs produced
 - In case both give different results, results obtained by practical tests are accepted as valid.

Alignment Test

- For alignment tests, the machine must be set up in the principal horizontal and practical planes and axes are checked with spirit levels, etc.
- Alignment tests include:
 - Static tests:
 - Checks the alignment accuracy of various parts of a machine tool.
 - Dynamic tests:
 - Checks for vibration and deflection of various machine parts under actual dynamic loading
 - Alignment tests are carried out in the loaded condition
 - Various parts are excited at working frequency and behavior of machine observed.

Performance Tests

- Performance tests are carried out with only finishing cuts in test conditions specified by the manufacturer
- They include tests for
 - Variations in rigidity of m/c tool components
 - Quality of manufacture
 - Influence of m/c fixtures, cutting tool, work-piece & system rigidity on m/c ing accuracy

Common Geometric Tests

- Levelling of machine tool in the horizontal & vertical plane
- Flatness of machine bed
- Straightness & parallelism of bed ways & bearing surfaces
- Perpendicularity of guideways & bearing surfaces
- True running of machine spindle & its axial movement
- Parallelism of spindle axis to guideways
- Line of movement fro cross slides, slides & carriage

Measures of Geometric Accuracy

- Straightness

- A line is said to be straight when the deviation of various points from two mutually perpendicular lines remains within stipulated limits
- Tolerance on the straightness of a line is the maximum deviation of the spread of points on either side of the reference line

Measures of Geometric Accuracy

- Flatness error is defined as
 - Minimum separation of a pair of parallel planes that will just contain all the points on the surface
 - Deviation of surface from best fit plane
- Parallelism of lines and planes:
 - Two planes / lines that do not share a point are said to be parallel

Measures of Geometric Accuracy

- Squareness of straight lines and planes:
 - 2 planes, lines or a straight line & plane are said to be square when error in parallelism in relation to a standard square does not exceed a limiting value.
- Roundness:
 - Out of roundness is the radial distance between the minimum circumscribing circle & maximum inscribing circle.

Measures of Geometric Accuracy

- Cylindricity:
 - Radial separation between two coaxial cylinders fitted to the total measured surface under test in such a way that their radial difference is minimum
- Runout:
 - Measures the trueness of running of a body about its own axis

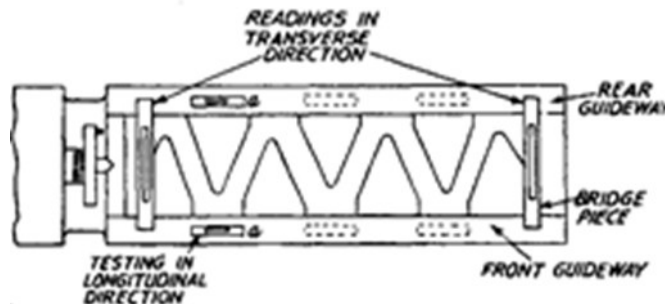
Acceptance Tests on Lathe

Leveling of the Machine

- First test to be carried out.
- In horizontal plane, tests are carried out in:
 - Longitudinal directions: Error will cause bed deflection, resulting in a simple bend & undesirable stresses.
 - Transverse direction: Errors will result in a twist. Thus the movement of the saddle will not be in a straight line.
- Front guide way should be convex only as the cutting forces and the weight of carriage act downward on it.
- Rear guide ways must be concave because of the tendency of the carriage to lift upwards, from the rear, under cutting forces.
- Transverse level may be in any direction, but no twist can be tolerated

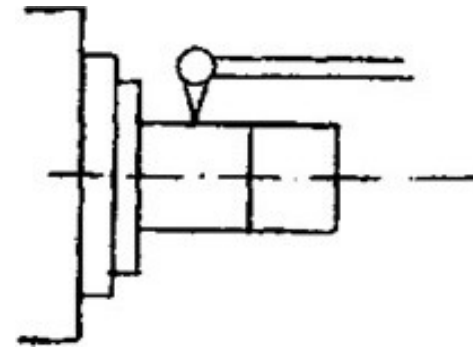
Leveling of the Machine

- Procedure:
 - Tested by a sensitive spirit level.
 - During the test the saddle is kept approximately in the center of the bed support feet.
 - Longitudinal direction:
 - Spirit level is traversed along the length of bed and readings at various places noted down.
 - It may be done by other methods, e.g., using straight edges, autocollimators or by taut wire method.
 - Transverse direction:
 - Spirit level is placed on a bridge piece to span the front and rear guide ways and then reading is noted.



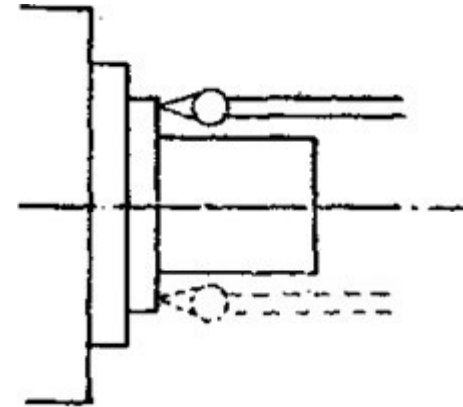
True Running of Locating Cylinder of Main Spindle.

- Locating cylinder locates the chuck or face plate.
- Locating surface is cylindrical and this must run truly; for only then the face plate etc., can run truly.
- Procedure:
 - The dial indicator is fixed to the carriage and the feeler touches the locating surface.
 - The surface is then rotated on its axis and indicator should not show any movement of needle.



Axial Slip of Main Spindle

- ❖ Axial play means the indispensable freedom of spindle movement in axial direction to prevent it from seizing by heating.
- ❖ Axial slip is defined as the axial spindle movement which follows the same pattern and is due to the manufacturing error.
- ❖ Due to axial slip, in screw cutting, the pitch will not be uniform.
- ❖ Not important while turning.
- ❖ Procedure:
 - The dial gauge is clamped to the bed & its feeler rests on the face of the locating spindle shoulder.
 - The locating cylinder is rotated & changes in reading noted down.
 - Readings are taken at two diametrically opposite points.
 - Total error (as indicated by the dial gauge) includes errors due to:
 - Axial slip due to error in bearings supporting the locating shoulder.
 - Face of the locating shoulder not in a plane perpendicular to axis of rotation.
 - Irregularities of front face.



True Running of Headstock Centre

- Tests the trueness of rotation of headstock center with the axis of rotation of the spindle.
- Errors will result in eccentricity while turning a work
- Procedure:
 - The feeler of the dial indicator is pressed perpendicular to the taper surface of the center, and the spindle is rotated.
 - Deviation indicated by the dial gauge gives the trueness of the center.

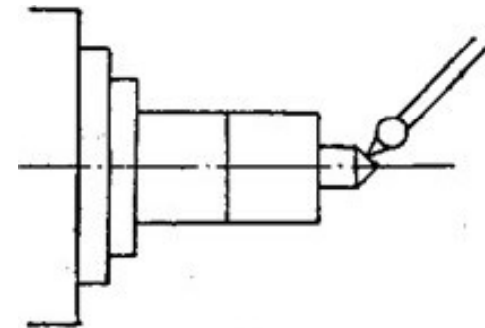
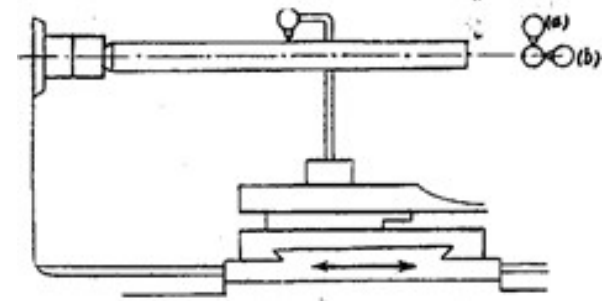


Fig. 16.4

Parallelism of Main Spindle & Saddle Movement

- Checked in both vertical and horizontal planes.
- Lack of parallelism will result in:
 - Horizontal direction: A tapered surface is produced.
 - Vertical direction: A hyperboloid surface is produced
- Procedure:
 - A mandrel is fitted in the taper socket of the spindle.
 - Requirements of mandrel:
 - It should have a concentric taper shank which is close fit to the spindle nose taper.
 - It must be so proportioned that its overhang does not produce appreciable sag
 - Dial indicator is mounted on the carriage & its feeler is pressed on the mandrel & the carriage is moved
 - Deviations in the horizontal plane are given by dial (b) and in vertical plane by dial (a).
 - In vertical plane the mandrel should be rising towards the free end in order to counteract the weight of mandrel and job. But for counter-acting cutting forces, it should be lower towards free end.
 - In horizontal plane, mandrel should be inclined in a direction opposite to the direction of tool pressure.



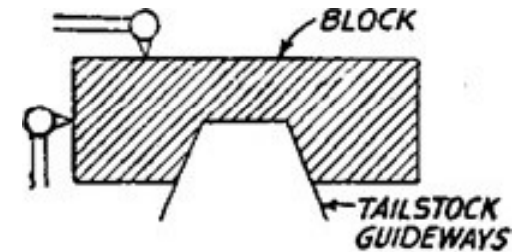
True Running of Taper Socket in Main Spindle

- Errors will result in production of eccentric and tapered jobs
- Procedure:
 - A mandrel is fitted into the tapered hole and readings at two extremes of the mandrel are taken by means of a dial indicator as shown.



Parallelism of Tailstock Guide-ways with Carriage Movement

- When the job is held between head-stock and tail stock centre for turning., its axis must coincide with the tailstock centre.
- If the tailstock guideways are not parallel with the carriage movement there a taper turning will result.
- Procedure:
 - The dial indicator is held in the carriage
 - A block is placed on the guideways as shown and the dial indicator feeler is touched on the horizontal and vertical surfaces of the block.
 - When the carriage is moved, errors are indicated by the pointer of dial indicator.



Movement of Upper Slide Parallel to Main Spindle

- Procedure:
 - A dial indicator is fixed in the tool post & a mandrel in the spindle.
 - The feeler of the dial gauge is pressed against the mandrel in vertical plane and the upper slide is moved longitudinally.

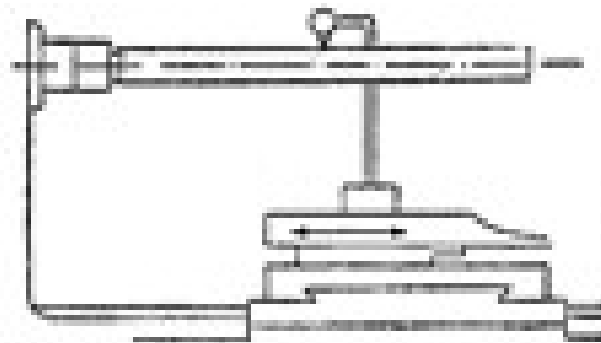


Fig. 16.8

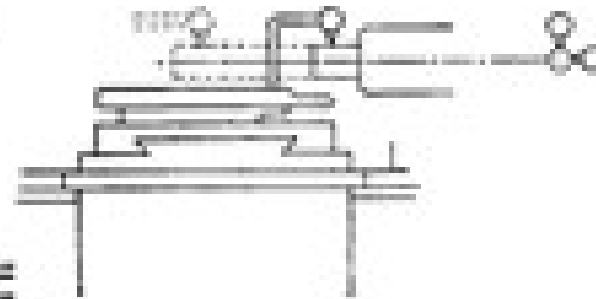
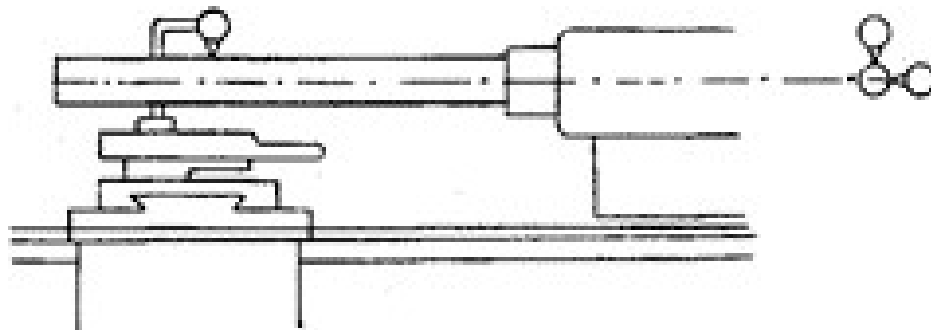


Fig. 16.9

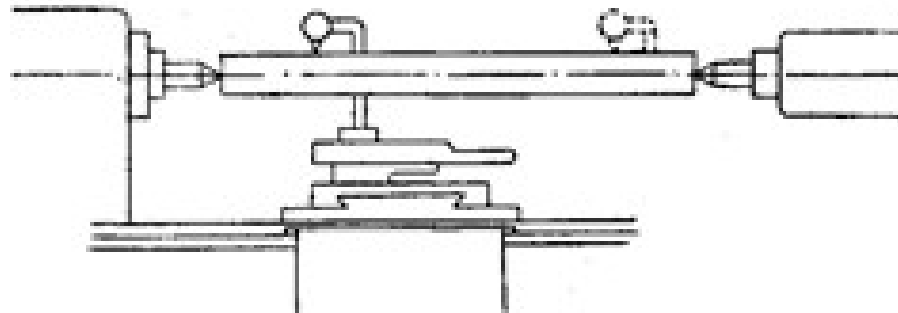
Parallelism of Tailstock Sleeve to Saddle Movement

- Errors in parallelism will result in the job being tilted up or down or in sideways when it is held between the centers.
- Tailstock sleeve should be rising towards the free end in vertical plane and should be inclined towards the tool pressure in horizontal plane.
- Procedure
 - Dial indicator is fixed on the tool post and its plunger is pressed against the sleeves first in vertical and then in horizontal plane.
 - The carriage is moved along the full length of the sleeve and deviations as indicated by dial indicator are noted down.



Alignment of Centers in Vertical Plane

- If the axes are not parallel to each other, axis of a job is fitted between the centers will not be parallel to the carriage movement.
- This test is to be carried out in vertical plane only.
- A mandrel is fitted between the two centers and dial gauge on the carriage. The feeler of the dial gauge is pressed against the mandrel in vertical plane as shown in Fig. 16.11 and the carriage is moved and the error noted down.



Pitch Accuracy of Lead Screw

- Accuracy of threads cut on any machine depends upon the accuracy of its lead screw.
- Thus the pitch of the lead screw throughout its length be uniform.
- Procedure
 - A positive stop is fixed to the lathe bed. Against the stop, the length bars and slip gauges can be located. An indicator is mounted on the carriage. it makes contact against the calculated length of slip gauges. The initial reading of the dial gauge against the slip gauge is noted.
 - The slip gauges are removed and the carriage is connected to the lead screw (disconnected from the gear train). An indexing arrangement rotates the lead screw so that distance travelled by carriage is equal to the length of slip gauges. The reading of the dial indicator against the stop is noted down in this position.
 - The two readings are compared. Any difference indicates an error.

Alignment Tests on Drilling Machine

Flatness of Clamping Surface

- To check that it is properly installed in both horizontal (longitudinal and transverse directions) and vertical directions
 - Flatness of clamping surface of base:
 - Straight edge is placed on two gauge blocks
 - Feeler gauges is inserted to measure error
 - Error $\leq 0.1/1000$ mm clamping surface
 - Surface should be concave only
 - Flatness of clamping surface of table
 - As above

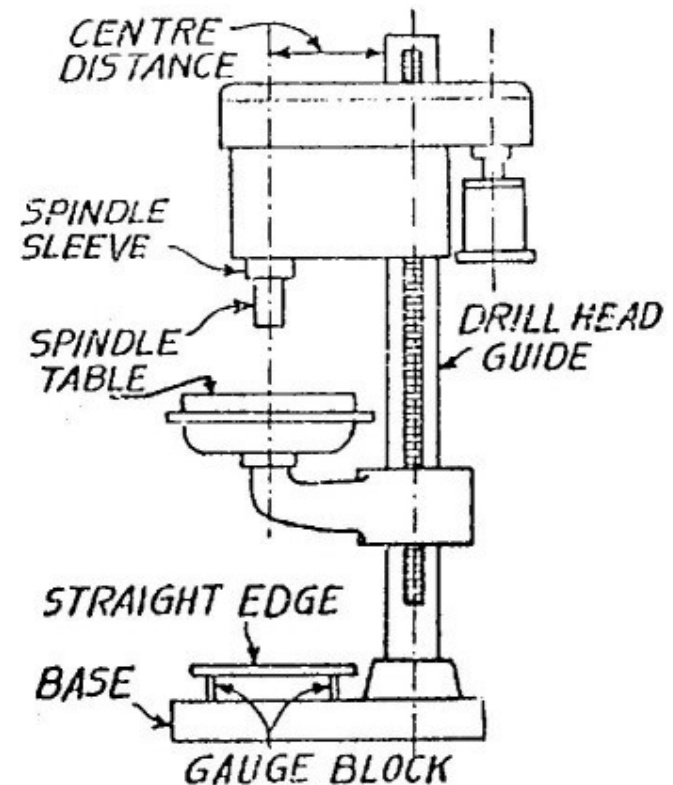


Fig. 16.17. Checking flatness of clamping surface of base.

- Perpendicularity of drill head guide to the base plate :
 - Tested in both vertical plane (a) and in a plane at 90^0 to the above plane (b)
 - Frame level with graduations from 0.03 to 0.05 mm/m
 - Error should not exceed 0.25/1000 mm for (a) and 0.15/1000 mm for (b)
- Perpendicularity of drill head guide with table :
- Perpendicularity of spindle sleeve with base plate :
 - Tested in both the planes using frame level
 - Error should not exceed 0.25/1000 mm for plane (a) and 0.15/1000 mm for plane (b)

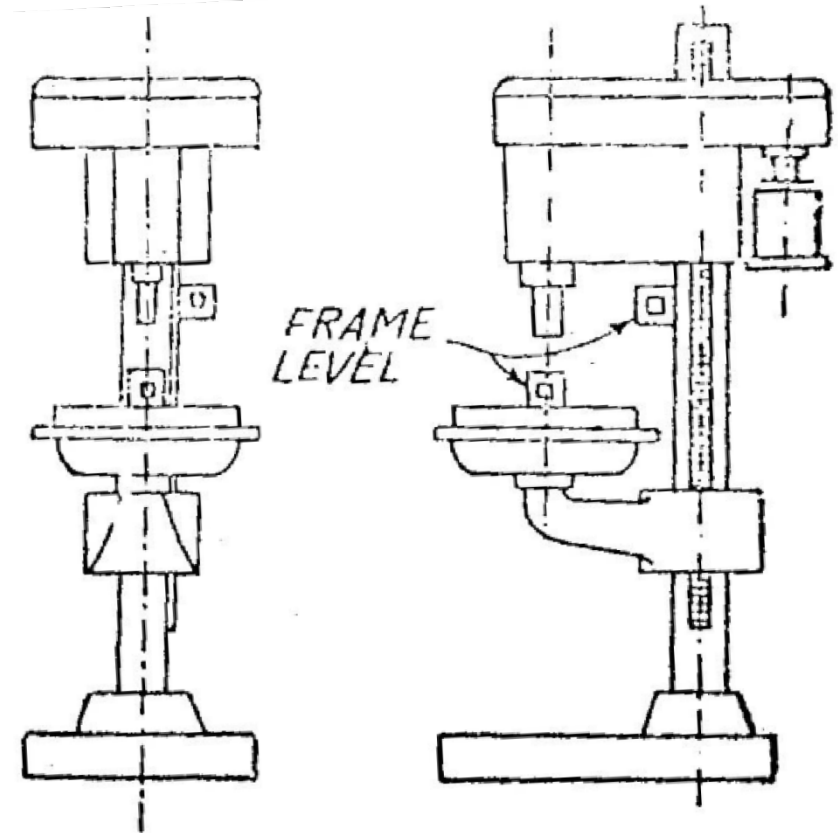
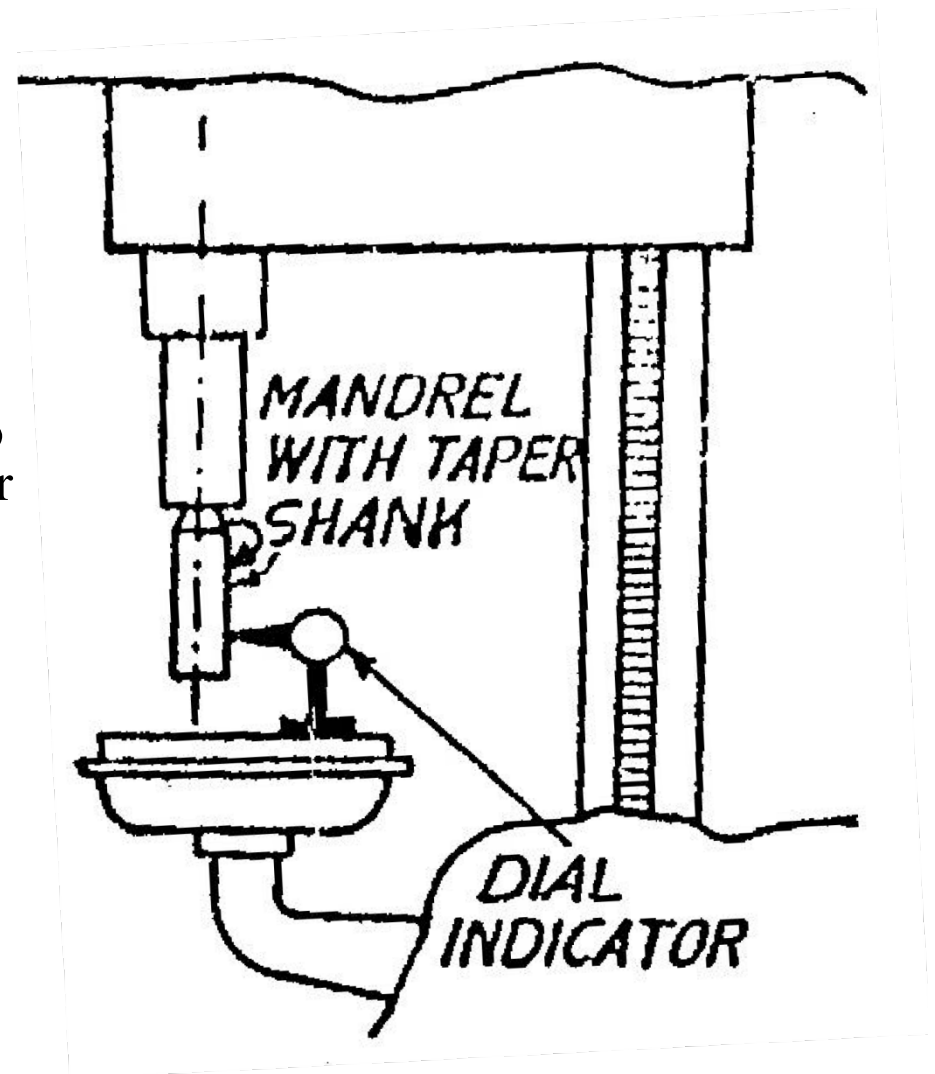


Fig. 16.18. Test for perpendicularity of drill head with table.

- True running of spindle taper :

- Test mandrel is placed in the tapered hole of spindle
- Dial indicator
- Spindle rotated slowly
- Error should not exceed 0.03/100 mm for machines with taper up to Morse No. 2 and 0.04/300 mm for machines with taper larger than Morse No. 2



- Parallelism of the spindle axis with its vertical movement :
 - Test is performed into two planes A & B
 - Test mandrel and dial indicator
 - Spindle is adjusted in the middle position of its travel
 - Permissible errors are : For machines with taper up to Morse No. 2 0.03/100 mm for plane A and 0.03/100 mm for plane B.
 - For machines with taper larger than Morse No. 2 0.05/300 mm for plane A and 0.05/300 mm for plane B.

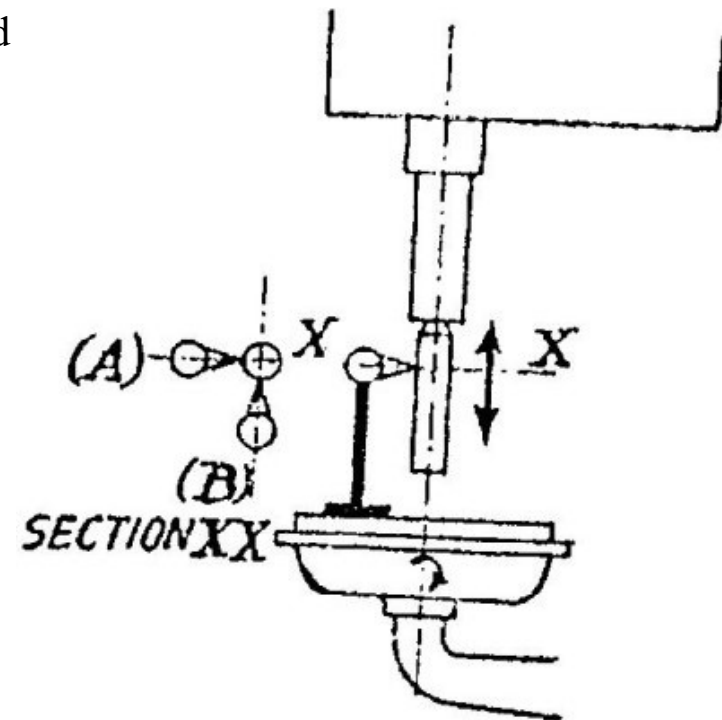
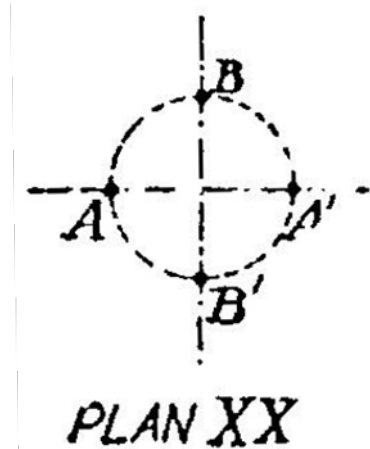
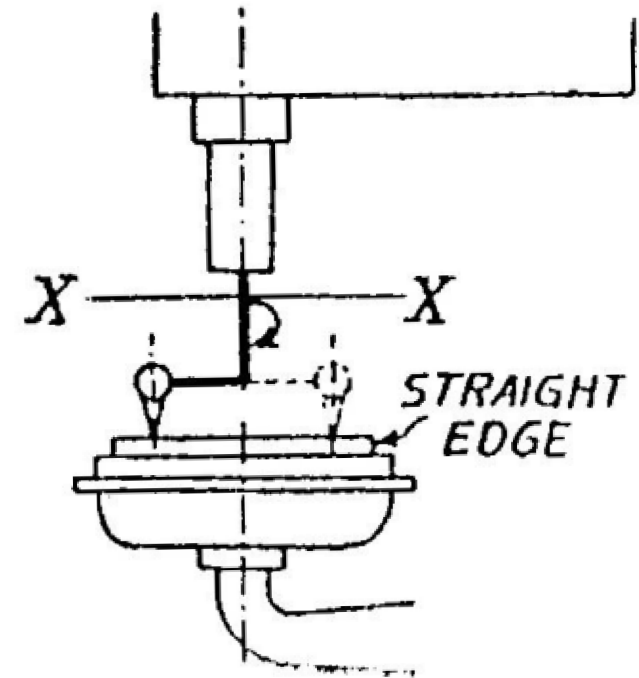


Fig. 16.20

- Squareness of spindle axis with table :
 - Straight edge placed in position AA' and BB'
 - Work table in middle position of its vertical travel
 - Dial indicator
 - $A - 1800 - A'$
 - The permissible errors are $0.08/300$ mm for AA' and $0.05/300$ mm for BB'.



- Total deflection :
 - Drill head and table are arranged in their middle position.
 - Dial indicator is used
 - Dynamometer or load gauge provides the weight

<i>Drilling Diameter in Open Hearth Steel</i>	<i>kg</i>
6	100
10	200
16	350
20	550
25	750
32	950
40	1200

	<i>for centre distance upto and including</i>
0.4 mm	200 mm
0.6 mm	200-300 mm
0.8 mm	300-400 mm
1.0 mm	400 and above mm

